
Qiskit-Based Simulation of HAP-Enabled Multi-User Entanglement QKD over FSO Channels

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Outline

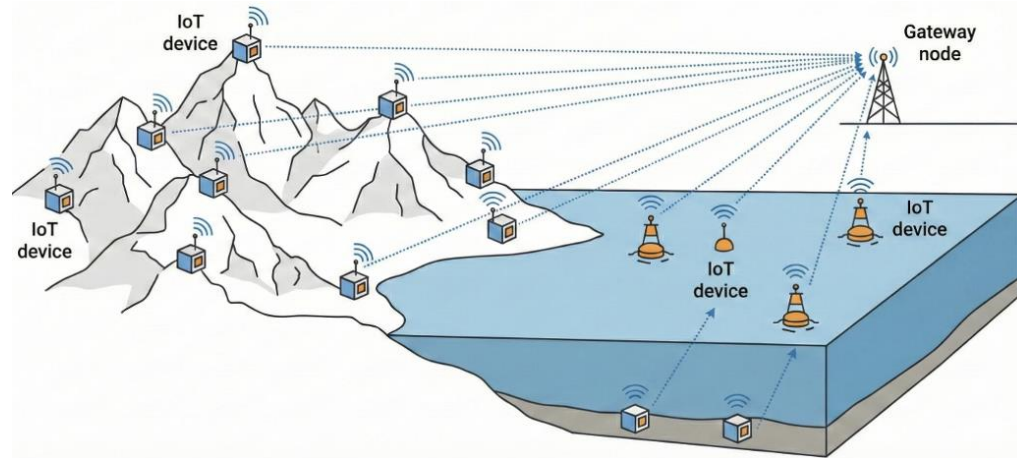
1. Introduction
2. System Description and Qiskit-based Simulation
3. Results and Discussions
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1. Introduction

Background

- Rapid IoT Growth: Connecting remote/resource-constrained devices.
- Deployment Challenges: Devices often operate in remote, resource-constrained areas, as shown in the figure on the right.
- The Threat of Quantum Computers: Classical cryptography is vulnerable to future quantum computer attacks.

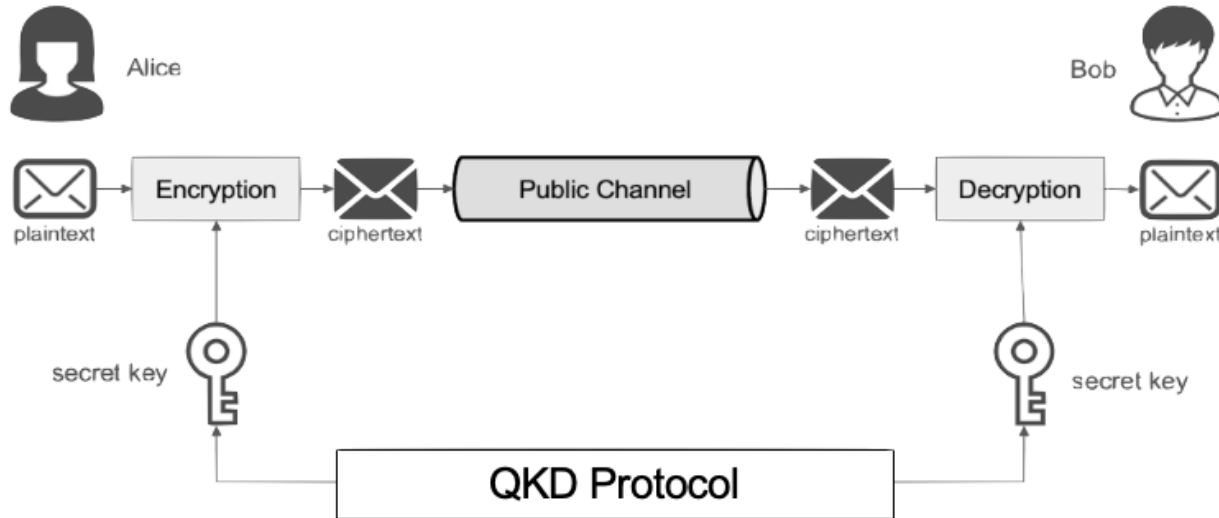
New security technologies are needed
→QKD is attracting attention



What is Quantum Key Distribution (QKD)

QKD is a secure communication technology to distribute secret keys.

- Based on quantum physics.
- Guarantees **unconditional security**



Research Motivation

- **Problem: Most of recent works for QKD over IoT networks have relied on optical fiber**
 - Not suitable for global QKD networks and massive IoT environment
- **Possible Solution: Using HAP-based FSO for QKD**
 - Advantage: can support for massive IoT environment, especially rural, oceanic, and remote areas
 - Low cost, flexible deployment to compare satellite-based QKD
- **Alternative to Optical Fiber:** HAP replace the need for physical optical fiber links.
- **Addressing Remote Area Challenges:** Overcomes the impracticality of laying fibers in rural areas (e.g., mountains, islands).
- **Scalability for Gateways:** Eliminates the need for extensive fiber infrastructure to numerous gateways.

Research Motivation

- **Research Questions:**

- (1) How HAP-based FSO can provide QKD services for secure IoT Networks?
- (2) What is the achievable performance?
- (3) What is an efficient QKD protocol to support multiple users

Research Goals

- (1) Propose the architecture with **HAP**-based FSO for QKD
 - HAP shares the secret key to multiple users via QKD
- (2) Develop a **Qiskit-based simulation framework** incorporating realistic FSO channel
 - Evaluate the **secret key rate**
- (3) Propose an efficient QKD protocols to support multiple users
 - HAP to multiple users: **BBM92 protocol with TDMA**

2. System Description and Qiskit-based Simulation

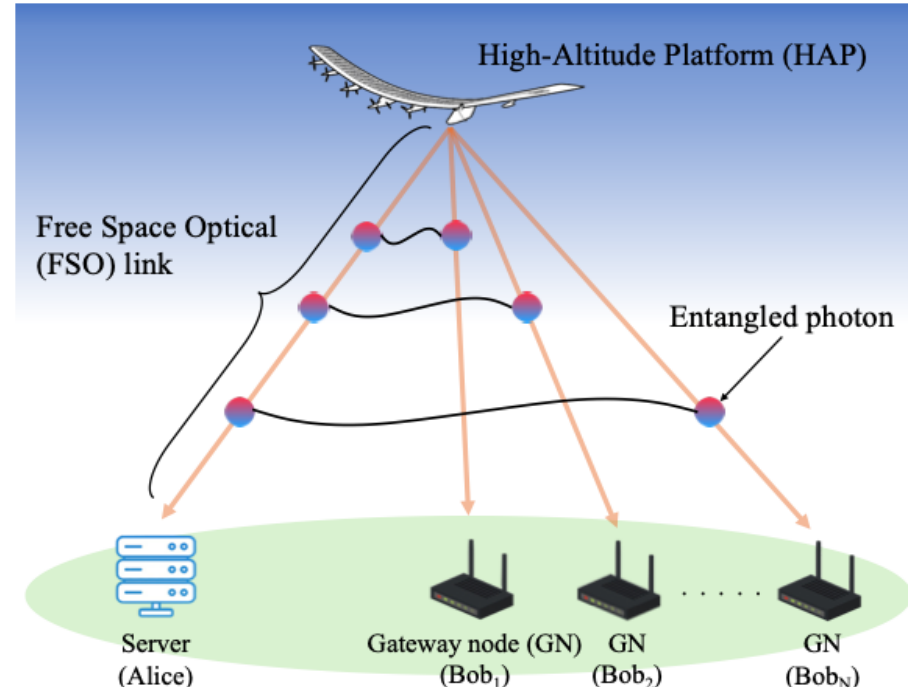
System Model

QKD Protocol: BBM92 for key generation and sharing.

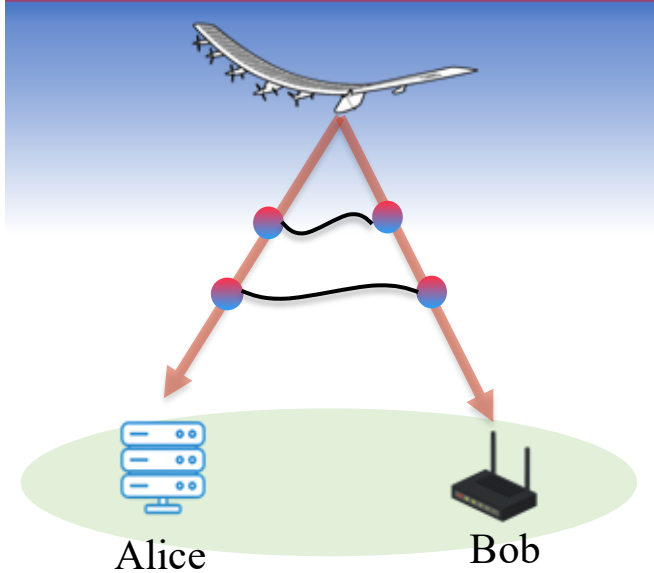
HAP: Located in the stratosphere. Generate and distribute entangled photon pairs to the ground.

Server: A central server always receives one photon from the HAP

Gateway Nodes: Multiple regional nodes for secure communication with the server. Receive the other photon from the pair via TDMA.



BBM92 Protocol



 Alice

Photon				
Basis				
Qubit				
Bit	1	0	1	1

 Bob

Photon				
Basis				
Qubit				
Bit	1	0	1	0

Basis \ Bit	0	1
		
		

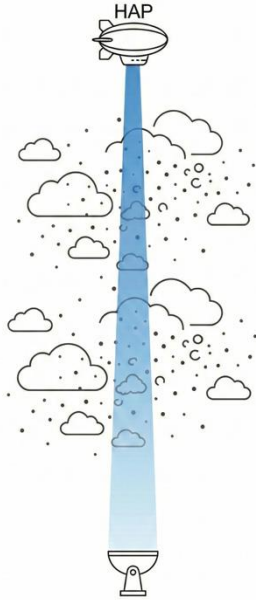
Basis Match				
Sifted key	1	None	1	None

 Record "11" as the sifted key

FSO Channel

Photons are transmitted through free-space optics (FSO) channel

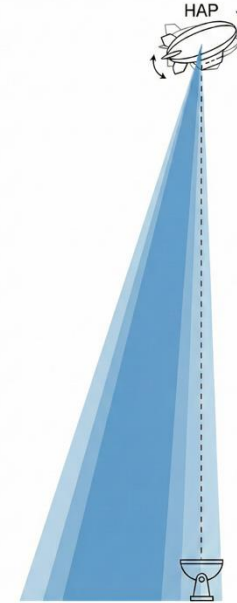
Atmospheric Attenuation



Atmospheric Turbulence



Beam Spreading Loss and Misalignment



These factors cause photon loss and photon state changes
→ Affect key generation

BBM92 protocol with TDMA (Time Division Multiple Access)

TDMA:

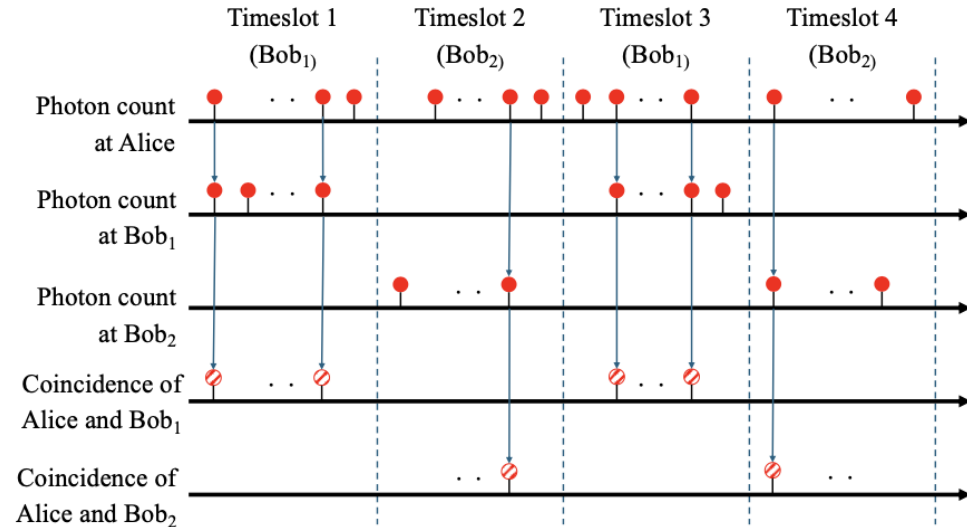
- Time is divided into fixed slots assigned to each Bobs

Sequential Beam Switching:

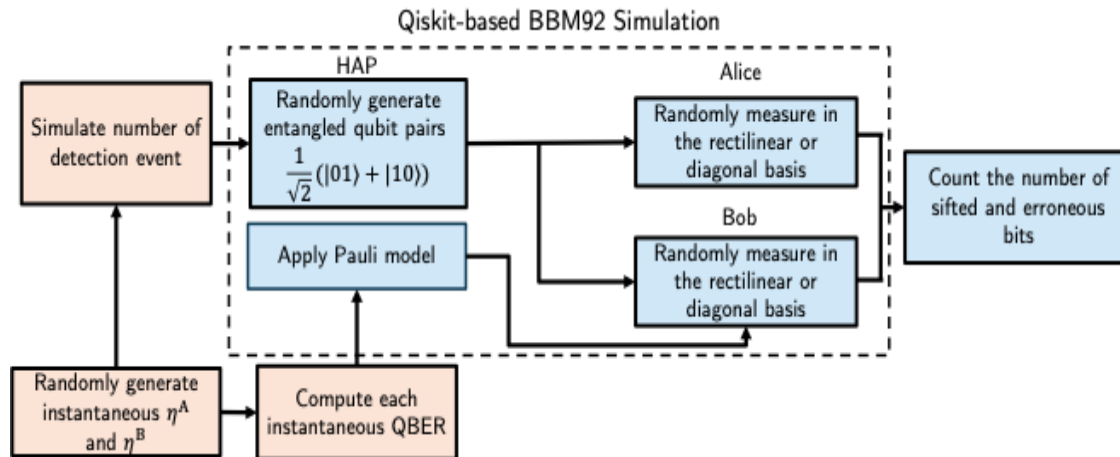
- To Alice: Always connected.
- To Bob_i: Scheduled Bob sequentially

Key Generation:

- Attempting key generation when **coincidence detection** between Alice and Bob.



Qiskit-based simulation framework



Implementation of the BBM92 Protocol Using **Qiskit**.

- **Qiskit**: An open-source SDK provided by IBM for quantum computing.

Integrating detection event into Qiskit:

The transmittance derived from the **FSO factors** determines the coincidence detection probability.

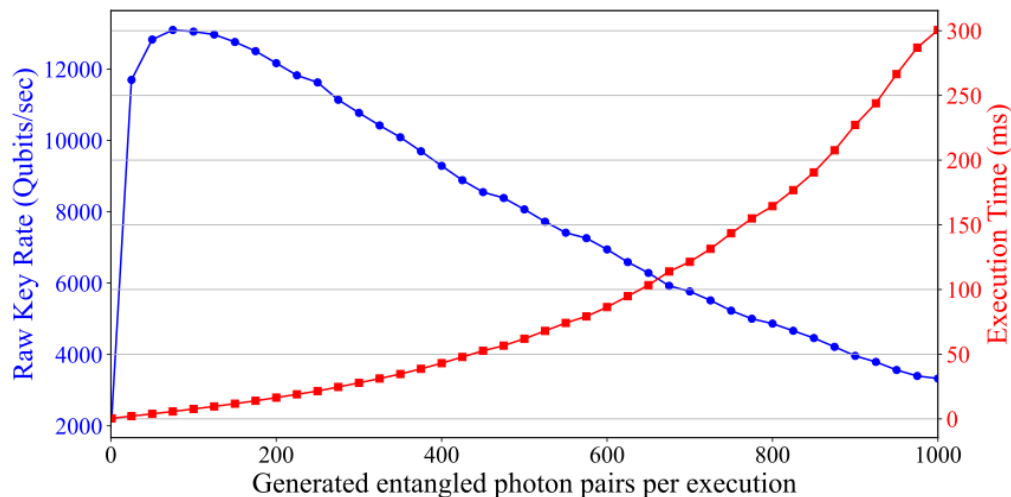
This number of coincidences is input into the **Qiskit-based simulation**

3. Results and Discussions

Performance of Simulation Framework

Raw Key Rate (qubits/sec) generated by Qiskit

- **Optimal Batch Size:**
The rate peaks when generating **75 pairs per execution**.
- **Max Throughput:**
Achieves a peak rate of **$\sim 13,000$ qubits/sec**.



Simulation Result: Achievable Secret Key Rate (SKR)

User Positions:

- **Alice:** Zenith Angle = 30°
- **Bob₁:** Zenith Angle = 10°
- **Bob₂:** Zenith Angle = 45°

Optimal Source (μ):

Average number of photon pairs generated per pulse.

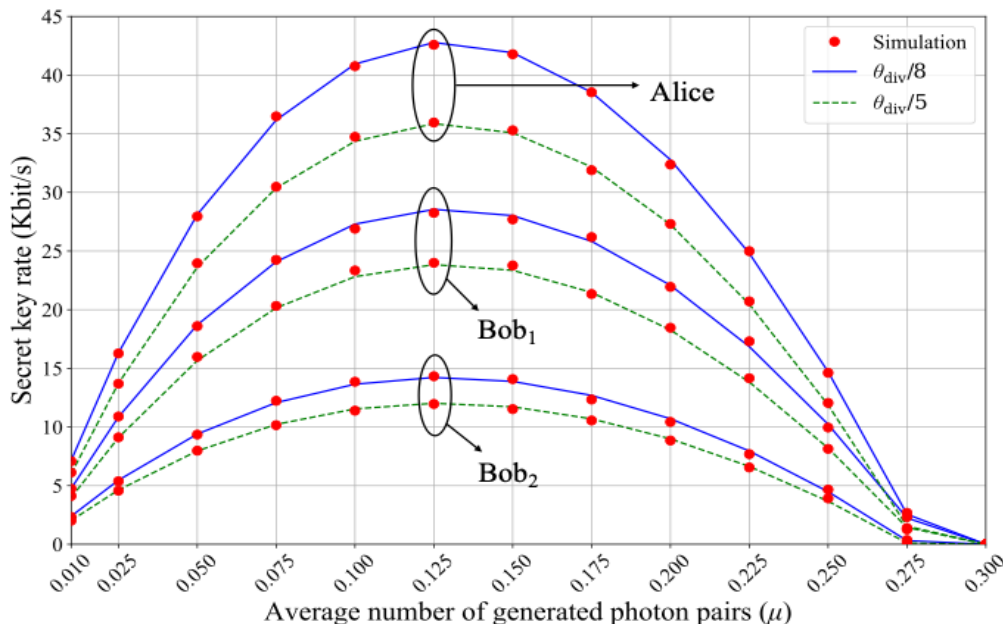
SKR peaks at $\mu = 0.125$.

Zenith Angle:

Bob₁ outperforms **Bob₂** due to lower atmospheric loss.

Beam Divergence (θ_{div}):

Smaller jitter ($\theta_{div}/8$) higher SKR than larger jitter ($\theta_{div}/5$).



Scalability Analysis: Number of Supported Users

Evaluate how many Bobs can be supported

Assume the HAP serving Alice and 5 Bobs

- **Alice** : Zenith Angle = 30°
Atmospheric transmittance
($\tau_{zen,A}$) = $0.1 \sim 0.9$
- **Bob_{1~5}**: Zenith Angle = $10^\circ, 20^\circ, 30^\circ, 40^\circ, 50^\circ$

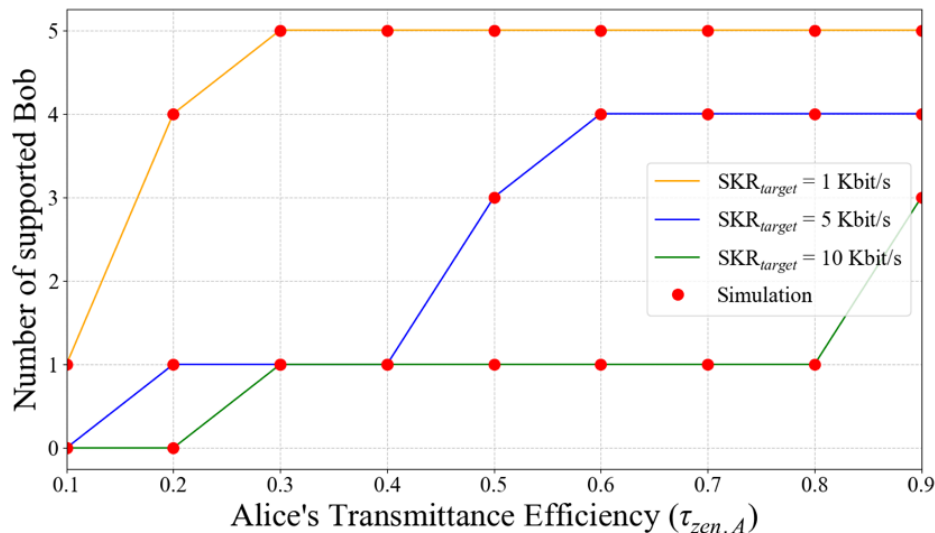
Supported:

Achievable **SKR** $\geq$ **Target SKR** (SKR_{target})

$SKR_{target} = 1Kbit/s$: Supports all 5 users even with moderate sky clarity ($\tau_{zen,A} \geq 0.3$).

$SKR_{target} = 5Kbit/s$: **Step-wise improvement**
($\tau_{zen,A} \geq 0.5$)

$SKR_{target} = 10Kbit/s$: Requires extremely clear sky conditions ($\tau_{zen,A} \geq 0.9$).



Conclusion

- Proposed a HAP-based FSO-QKD network using BBM92 protocol and TDMA.
- Developed a Qiskit-based simulation framework incorporating realistic FSO channel effects.
- Simulation results show strong consistency with theoretical models.
- Confirmed the feasibility for scalable, secure quantum communications of proposed architecture.